



Quintin de Jongh

Mechanical Engineer

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 Cape Town

Summary

Mechanical Project Engineer with 5+ years of experience spanning advanced product development, thermal-fluid R&D, industrial automation, and highly regulated plant utilities. Proven track record leading complex mechanical designs from concept, prototyping, and advanced simulation/optimization (CFD/FEA) through to mechatronic integration (PLC/HMI), hands-on fabrication, and full commissioning. Expert in transitioning innovative concepts into robust, manufacturable solutions for extreme environments, including high-temperature reactive flows and multi-billion dollar pharmaceutical infrastructure.

Professional Experience

Mechanical Project Engineer

January 2026 - Present

Calibre Life Sciences

- Lead the mechanical engineering and technical design and implementation for high-spec utilities within a multi-billion dollar pharmaceutical manufacturing facility, including the design of steam, chilled water, compressed air, potable water, water treatment and gas cylinder storage systems.
- Manage complex life sciences infrastructure projects, coordinating the end-to-end design and engineering of process drainage, thermal fluids, and chiller configurations to ensure strict compliance with international manufacturing standards.
- Develop comprehensive P&IDs, equipment schedules, and technical specifications, driving cross-functional project execution to deliver highly regulated mechanical utilities on tight engineering timelines.
- Dealt with diverse, large-scale industrial projects, including the engineering of plant utilities and HVAC systems for a major fruit juice producer, as well as executing complex CFD simulations to validate HVAC designs for a prominent pharmaceutical manufacturing facility in Port Elizabeth.

Mechanical Engineer - CAE and Product Development Consultant

October 2024 - January 2026

Cauchy Consult

- Led mechanical design and product development for advanced thermal-fluid energy systems, notably driving the optimization of a hydrogen reformer to enhance thermal efficiency, flow uniformity, and robustness in high-temperature environments.
- Executed complex CFD simulations (OpenFOAM) and 3D modeling (FreeCAD, SALOME) to guide design iterations, successfully translating advanced CAE insights into manufacturable components and comprehensive cross-functional client reports.

Mechanical Engineer

January 2023 - October 2024

Skeg Product Development

- Led the end-to-end design, commissioning, and experimental validation of a high-temperature hydrogen reformer through two prototype generations, and a Direct Contact Membrane Distillation desalination system deployed across Saudi Arabia.
- Driven by a hybrid execution model combining hands-on technical tasks (welding, fabrication, assembly) with advanced mathematical modeling, CFD/FEA simulations, budget oversight, and full-stack software development (TwinCAT 3 PLC, Angular UI).

Graduate Engineer

March 2022 - December 2022

BMEC Technologies

- Contributed to the development and commissioning of an ozone dispersion sanitation system and a zone-based automatic seed dispersion device by supporting mechanical and mechatronic design, Flutter web and micro-controller programming, and PCB design and assembly.

Education

MSc in Mechanical Engineering (with Distinction)

February 2021 - April 2022

University of Cape Town

Thesis: "Flexible Media Polishing Machine Design, Development and Experimentation for the Polishing of Ti-6Al-4V Components"

BSc in Mechanical Engineering

January 2016 - December 2020

University of Cape Town

Thesis: "Development of an Intelligent Grinding System", awarded with Distinction
Dean's Merit List for High Academic Achievement (2017)

Relevant Skills

- Product and Mechanical Design
- Computer Aided Engineering
- Machine Building
- Machine Control and Automation
- System Design
- Mechanical Simulations
- Process Modeling and Validation
- Experimental Design and Implementation (DOE)
- Project Management
- Front-End Web Development
- Client Liaison

Softwares and Packages

- SOLIDWORKS
- PTC Creo Parametric and Simulate
- OpenFOAM and Autodesk Simulation CFD
- FreeCAD
- SALOME and Paraview
- BECKHOFF TwinCAT 3 (PLC and HMI)
- MATLAB
- HTML, CSS, JS, and Flutter/Dart
- Python
- 3D Printing Slicer Softwares
- MS Office (including Visio & Project)
- Adobe Softwares (InDesign, Illustrator)

Journal Papers and Academic Experience

- During my time in academia (postgraduate), I was published in three journal papers and three conference papers relating to advanced manufacturing techniques.
- I worked as a lecturer for a year while completing my Master's degree and I also convened and worked as a primary lecturer for one course (manufacturing sciences for engineers).
- For full details of my academic experience and journal history, please visit my self coded portfolio website's curriculum vitae page: <https://quino97.github.io/contact.html>

Hobbies and Interests

Software Development

- Intermediate Python & Dart/Flutter
- Front-End Web Development using HTML, CSS & JS

Running and Fitness

- Completed over 20 races (including 2 marathons)
- A 5km Personal Best time of 17:30 and a half-marathon Personal Best of 1:21:53

Woodwork, 3D Printing and Household Projects

- I enjoy 3D printing gadgets and mechanical devices (FDM and Resin) as well as woodworking projects and home improvement e.g. building outdoor couches and a Murphy bed.